Quantitative Methods for Management

Tutorial Exercise Sheet: Applications to Dynamical Systems

Professor: **Fausto Gozzi**
Teaching Assistants: **Fabian Fuchs** and **Michele Ricciardi**

Recap: Solving a Discrete Dynamical System Using Eigenvectors. Consider

$$\mathbf{x}(t+1) = \mathbf{A}\mathbf{x}(t), \qquad t = 0, 1, 2, \dots.$$

Suppose the $n \times n$ matrix $\mathbf{A}$ has n linearly independent real eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_n$, with corresponding real eigenvalues $\lambda_1, \dots, \lambda_n$. Then the general solution is

$$\mathbf{x}(t) = \sum_{i=1}^{n} c_i \lambda_i^t \mathbf{v}_i,$$

where the constants $c_i \in \mathbb{R}$ do not depend on t. At $t = 0$, every factor λ_i^0 is taken as 1, including when $\lambda_i = 0$.

Each term satisfies the update equation because

$$\mathbf{A}\big(c_i \lambda_i^t \mathbf{v}_i\big) = c_i \lambda_i^{t+1} \mathbf{v}_i.$$

The eigenvectors form a basis, so every initial state can be represented this way. Along each eigenvector, the state is multiplied by the corresponding eigenvalue at every step.

- **Find the eigenvalues and eigenvectors.** Solve $\det(\mathbf{A} - \lambda I) = 0$ and find a basis of eigenvectors. The formula above requires n linearly independent eigenvectors!
- **Write the general solution.** Associate each eigenvector with the power of its corresponding eigenvalue.
- **Determine the constants.** For an initial condition, solve $\mathbf{x}(0) = \sum_{i=1}^{n} c_i \mathbf{v}_i$. For a condition at a later integer time T, solve $\mathbf{x}(T) = \sum_{i=1}^{n} c_i \lambda_i^T \mathbf{v}_i$. If every eigenvalue is nonzero, a prescribed state at time T determines the constants uniquely.
- **Check additional conditions.** Write the components explicitly. The condition $\mathbf{x}(t) \geq 0$ means that every component is nonnegative at every required time.
- **Study the limit.** Terms with $|\lambda_i| < 1$ tend to zero, terms with $\lambda_i = 1$ remain constant, terms with $|\lambda_i| > 1$ diverge, and terms with $\lambda_i = -1$ do not converge.

Exercise 1 (General solutions and conditions on the state)**.** Consider

$$x_1(t+1) = 2x_1(t) - x_2(t),$$
$$x_2(t+1) = -x_1(t) + 2x_2(t),$$

(a) Write the system as $\mathbf{x}(t+1) = \mathbf{A}\mathbf{x}(t)$. Find the eigenvalues of A and a corresponding basis of eigenvectors.

(b) Use your results to write the general solution in the form

$$\mathbf{x}(t) = c_1\lambda_1^t\mathbf{v}_1 + c_2\lambda_2^t\mathbf{v}_2.$$

Also write explicit formulas for $x_1(t)$ and $x_2(t)$.

(c) Find the unique solution satisfying $\mathbf{x}(0) = (1, 1)^T$.

(d) Find the unique solution satisfying $\mathbf{x}(4) = (2, 0)^T$.

(e) Determine all solutions satisfying $\mathbf{x}(t) \geq 0$ componentwise for every integer $t \geq 0$.

(f) Determine all solutions for which a finite limit $\lim_{t\to\infty} \mathbf{x}(t)$ exists, and find that limit.

Recap: Dynamical Systems and the Bullwhip Effect. The *Bullwhip effect* describes the amplification of, e.g., customer-demand fluctuations in orders placed with suppliers. Inventory corrections and delayed adjustments can cause orders to overshoot and then fall below their normal level. A discrete dynamical system describes how a state changes from one period to the next:

$$\mathbf{x}_{t+1} = \mathbf{A}\mathbf{x}_t, \qquad \mathbf{x}_t = \mathbf{A}^t\mathbf{x}_0.$$

Here, $\mathbf{x}_0$ is the initial state. When the entries represent deviations from target values, $\mathbf{x} = 0$ represents equilibrium. The eigenvalues of $\mathbf{A}$ help us determine how an initial disturbance develops.

For a real 2×2 matrix with nonreal eigenvalues $\lambda_1 = a + bi$ and $\lambda_2 = a - bi$, where $b \neq 0$, both eigenvalues have the same modulus

$$r = |\lambda_1| = |\lambda_2| = \sqrt{a^2 + b^2}.$$

The nonreal eigenvalues produce oscillations, while their modulus determines whether the amplitude shrinks or grows.

- **Find the eigenvalues.** Solve $\det(\mathbf{A} - \lambda I) = 0$.
- **Check their modulus.** For a nonreal conjugate pair, $r < 1$ gives shrinking oscillations, $r = 1$ gives persistent oscillations, and $r > 1$ gives growing oscillations.
- **Interpret the result.** Shrinking oscillations return the system toward equilibrium. Growing oscillations amplify a nonzero initial disturbance. Individual entries need not increase in absolute value at every step.

Growing oscillations provide one simple model of the bullwhip effect. However, temporary amplification of orders can also occur in a system that eventually returns to equilibrium.

Exercise 2 (Inventory adjustments and the bullwhip effect)**.** A retailer normally sells and orders 100 units per week. A one-time increase in demand leaves inventory 10 units below its target. Demand then returns to 100 units per week.

Let I_t denote inventory minus its target and O_t denote the order minus 100. Assume

$$I_{t+1} = I_t + O_t, \qquad O_{t+1} = \alpha I_t + \beta O_t, \qquad \begin{pmatrix} I_0 \\ O_0 \end{pmatrix} = \begin{pmatrix} -10 \\ 0 \end{pmatrix}.$$

The first equation assumes that the order placed at time t contributes to inventory over the following period.

(a) Explain why $\alpha < 0$ provides a corrective response to an inventory shortage. What does $0 \le \beta \le 1$ mean for the influence of the current order on the next order?

(b) Set $\alpha = -4$ and $\beta = 1$ for parts (b)–(e). Write the system as $\mathbf{x}_{t+1} = \mathbf{A}\mathbf{x}_t$, defining both $\mathbf{x}_t$ and $\mathbf{A}$.

(c) Calculate the eigenvalues and a corresponding eigenvector for each. Find their common modulus and determine whether the oscillations shrink or grow.

(d) Starting from $I_0 = -10$ and $O_0 = 0$, calculate the inventory and order deviations after one, two, and three weeks. Determine the actual orders. Which order deviations exceed the original demand shock in absolute value?

(e) Explain the initial increase in orders and the subsequent drop below normal. Use the eigenvalues to predict the subsequent behavior. Relate your findings to the bullwhip effect.

(f) Now set $\beta = \frac{1}{2}$ and let $\alpha < -\frac{1}{16}$, which ensures that the eigenvalues are nonreal. Using the fact that their product equals $\det(\mathbf{A})$, express their common modulus in terms of α. For which values of α do the oscillations shrink, persist, or grow? Interpret the result in terms of the strength of inventory corrections.

Recap: Weighted Graphs and Transition Matrices. A directed graph consists of vertices and arrows connecting them. Each arrow can carry a nonnegative weight representing the relative frequency of that transition. A loop represents remaining at the same vertex. The weighted adjacency matrix $\mathbf{A}$ records these weights using the convention

$$A_{ij} = \text{weight of the arrow from vertex } j \text{ to vertex } i,$$

with $A_{ij} = 0$ when there is no arrow. Thus, column j describes the arrows leaving vertex j, and

$$d_j = \sum_i A_{ij}$$

is their total weight. Giving every arrow weight 1 recovers the ordinary adjacency matrix.

Suppose each transition is chosen with probability proportional to its weight, depending only on the current vertex. We obtain the Markov transition matrix $\mathbf{P}$ by normalizing each column:

$$P_{ij} = \frac{A_{ij}}{d_j}, \qquad d_j > 0.$$

Here, P_{ij} is the probability of moving from j to i in one step. All entries are nonnegative, and every column sums to 1.

- **Build the weighted adjacency matrix.** Choose an order for the vertices and enter each arrow's weight in its destination row and starting column. Enter loop weights on the diagonal.
- **Normalize the columns.** Divide each column by its total weight. If that total is zero, we let the traveler stay at that vertex: set its diagonal entry in $\mathbf{P}$ to 1 and all other entries in that column to 0.
- **Update the probabilities.** Let $\mathbf{p}^{(t)}$ be the column vector of probabilities after t steps, with nonnegative entries summing to 1. Then

$$\mathbf{p}^{(t+1)} = \mathbf{P}\mathbf{p}^{(t)}, \qquad \mathbf{p}^{(t)} = \mathbf{P}^t\mathbf{p}^{(0)}.$$

The entry $(\mathbf{P}^t)_{ij}$ is the probability of reaching i after t steps when starting at j.

- **Interpret paths.** Multiply the transition probabilities along a path. To find the probability of reaching a given destination, add the probabilities of all paths of the required length that end there. In particular,

$$(\mathbf{P}^2)_{ij} = \sum_k P_{ik}P_{kj}.$$

Recap: The Perron–Frobenius theorem and Stationary Distributions. Let $\mathbf{P}$ be a transition matrix with nonnegative entries and columns summing to 1. A probability distribution evolves according to

$$\mathbf{p}^{(t+1)} = \mathbf{P}\mathbf{p}^{(t)}.$$

A stationary distribution is a probability vector $\mathbf{p}^*$ satisfying

$$\mathbf{P}\mathbf{p}^* = \mathbf{p}^*, \qquad p_i^* \geq 0, \qquad \sum_i p_i^* = 1.$$

Thus, a stationary distribution is an eigenvector for the eigenvalue 1, with nonnegative entries normalized to sum to 1. Once this distribution is reached, it remains unchanged.

If every entry of $\mathbf{P}$ is strictly positive, the Perron–Frobenius theorem guarantees a unique stationary distribution, with strictly positive entries. The eigenvalue 1 occurs only once in the characteristic polynomial, and every other eigenvalue has modulus less than 1. Consequently,

$$\mathbf{P}^t\mathbf{p}^{(0)} \longrightarrow \mathbf{p}^* \qquad \text{as } t \to \infty$$

for every initial probability distribution $\mathbf{p}^{(0)}$.

- **Check the assumptions.** Verify that every column sums to 1 and every entry is strictly positive.

- **Find the eigenspace for the eigenvalue** 1**.** Solve $(\mathbf{P} - I)\mathbf{v} = 0$ to obtain

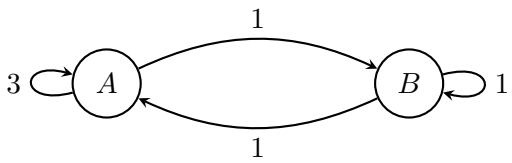
$$E_1 = \{\mathbf{v} : \mathbf{P}\mathbf{v} = \mathbf{v}\}.$$

  Under the theorem's assumptions, this eigenspace is one-dimensional and contains an eigenvector with strictly positive entries. Choose such a vector $\mathbf{v}$ and normalize it:

$$\mathbf{p}^* = \frac{\mathbf{v}}{\sum_i v_i}.$$

- **Interpret the result.** The entries of $\mathbf{p}^*$ give the long-run proportions, independently of the initial distribution.

Exercise 3 (Customer switching and long-run market shares)**.** Customers choose between brands A and B. The weighted graph below describes their monthly choices. The weights represent relative switching frequencies: among customers currently using A, staying is three times as common as switching; customers using B are equally likely to stay or switch.



Let $\mathbf{p}^{(t)} = (a_t, b_t)^T$ contain the two brands' market shares in month t.

(a) Construct the weighted adjacency matrix $\mathbf{A}$.

(b) Normalize its columns to obtain the transition matrix $\mathbf{P}$. Interpret its entries and verify that each column sums to 1.

(c) Initially, every customer uses brand A. Calculate the market shares after one and two months.

(d) Calculate $\mathbf{P}^2$. Interpret its entry in row B, column A, using the two possible paths from A to B over two months.

(e) Verify that the Perron–Frobenius theorem applies. Find the eigenspace for eigenvalue 1, normalize a positive eigenvector, and interpret the resulting stationary distribution.

(f) Find the other eigenvalue. Explain how it agrees with convergence toward the stationary distribution and compare the distributions from part (c) with that limit.